Data Analysis

Syllabus

Yu-You Liou

Fall 2025

Lecturer: Yu-You Liou	**Contact**: d10627008@ntu.edu.tw
Course Code: EIB-301-01-A1	**Website**: https://yyliou.github.io/da
Meeting Time: Tuesday, Periods 8–9 (15:10–17:00)	**Location**: A207

Course Objectives

1. To enable students to understand the basic logic, applications, and simple techniques of data analysis using R.
2. To develop students' ability to build clear and informative data visualizations grounded in the grammar of graphics.
3. To communicate data-driven findings clearly to both technical and non-technical audiences.
4. To integrate generative AI to enhance students' workflows in programming and compilation tasks.

Textbooks

1. Wickham, H., Navarro, D., & Pedersen, T. L. *ggplot2: Elegant Graphics for Data Analysis* (3rd ed.). (Main) https://ggplot2-book.org
2. Adler, J. *R in a Nutshell: A Desktop Quick Reference* (2nd ed.). O'Reilly Media.

Course Structure

1. **Lecture** (First hour): Foundational knowledge and concepts introduced each session.
2. **Lab** (Second hour): Hands-on in-class exercises using R to apply the concepts learned during the lecture.

Assessment Criteria

1. **Attendance** (12%): $R = \min\{12, t\}$, where t represents the number of sessions attended.
2. **In-Class Exercises** (48%): 12 exercises $\times$ 4% each, submitted at the end of each session.
3. **Mid-Term Oral Presentaiton** (20%) and **Final Oral Presentaiton** (20%).

Regulations

1. **Attendance**

Per Shih Chien University regulations, students who are absent for more than one-third of class sessions will receive a final grade of zero. Make-up attendance will not be permitted.

2. **Presentations**

Students who do not show up for their scheduled presentation will receive a grade of zero for that component. Presentation order will be posted on the course website.

3. **In-class exercise deadline**

The deadline for all in-class exercises is before midnight of the following week. Late submissions will not be accepted.

4. **Academic Integrity**

All submitted work must be the student's own. Copying from other students is strictly prohibited. AI detection tools are used to review submissions; any work found to be copied will receive a grade of zero.

5. **AI Policy**

Students may use AI tools (e.g., ChatGPT, GitHub Copilot) as learning aids, but must be able to explain and defend any code or analysis they submit. Uncritical submission of AI-generated work without understanding constitutes academic dishonesty.

Required Software

All software is free and open-source: **R** at https://cran.r-project.org, **RStudio Desktop** at https://posit.co/download/rstudio-desktop, and R packages `tidyverse`, `ggplot2`, `patchwork` (installation instructions on the course R page).

Course Schedule

1. Weeks 1–2: Part I — Introduction to R and first steps with `ggplot2`.
2. Weeks 3–7: Part II — Layers (individual geoms, collective geoms, statistical summaries, maps, networks, annotations, arranging plots).
3. Weeks 8–9: Mid-term presentations.
4. Weeks 10–11: Part III — Scales (position scales, colour scales, other aesthetics).
5. Weeks 12–14: Part IV — The grammar; advanced topics.
6. Weeks 15–16: Final presentations.
7. Weeks 17-18: Flexible learning weeks.

Syllabus is subject to change. Any updates will be announced on the course website.